Role of the RAS Pathway in Cancer

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Abstract

The RAS pathway, known as the Ras/Raf/MAPK pathway, is a crucial signal transduction cascade that regulates essential cellular processes, including growth, differentiation, and survival. Central to this pathway is the RAS protein, a small GTPase that acts as a molecular switch, alternating between inactive GDP-bound and active GTP-bound states. Activation of RAS is initiated by extracellular signals, primarily through the binding of ligands to receptor tyrosine kinases (RTKs), which subsequently activate guanine nucleotide exchange factors (GEFs) and the conversion of GDP to GTP. Notably, mutations in the RAS gene, especially the KRAS G12C mutation, are implicated in various cancers, particularly non-small cell lung cancer (NSCLC) and colorectal cancer (CRC), where they lead to uncontrolled cell proliferation and tumorigenesis. This study examines the intricate roles of KRAS, including its domains, isoforms, and mutation profiles, and discusses emerging therapeutic approaches targeting KRAS mutations, such as specific inhibitors like Adagrasib. A comprehensive understanding of the RAS pathways complexities and its oncogenic potential is essential for advancing targeted therapies and enhancing cancer treatment
outcomes.

Methodology

Several public databases and software tools were used to analyze the nucleotide, protein and gene expression data as shown below.

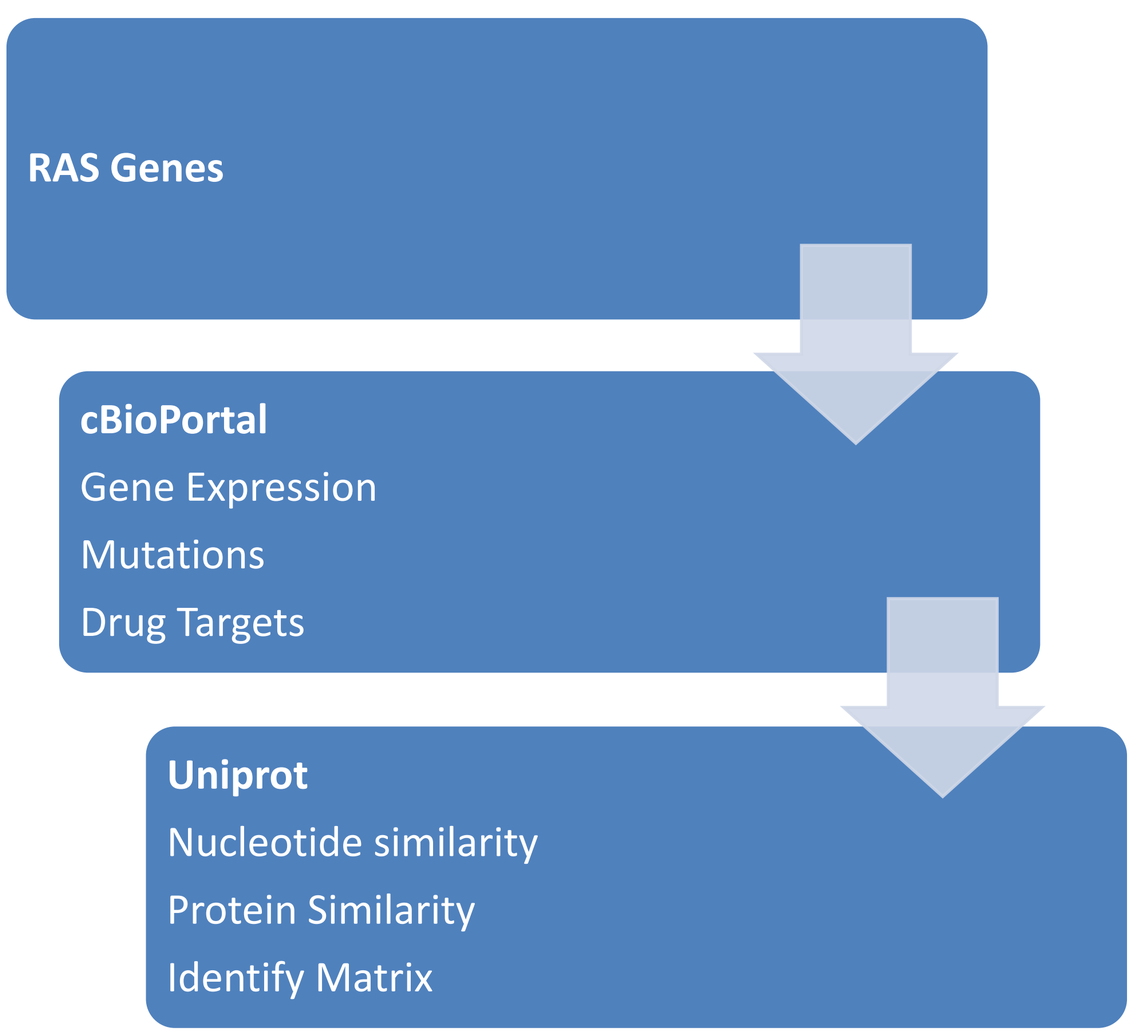

Figure 1 – RAS Pathway sequence and gene expression analysis workflow

Results

a)

Nucleotide Sequence (570 nt):

ATGACTGAATATAAACTTGTGGTAGTTGGAGCTGGTGGCGTAGGCAAGAGTGCCTTGACGATACAGCTAA
TTCAGAATCATTTTGTGGACGAATATGATCCAACAATAGAGGATTCCTACAGGAAGCAAGTAGTAATTGA
TGGAGAAACCTGTCTCTTGGATATTCTCGACACAGCAGGTCAAGAGGAGTACAGTGCAATGAGGGACCAG
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ACCATTATAGAGAACAAATTAAAAGAGTTAAGGACTCTGAAGATGTACCTATGGTCCTAGTAGGAAATAA
ATGTGATTTGCCTTCTAGAACAGTAGACACAAAACAGGCTCAGGACTTAGCAAGAAGTTATGGAATTCCT
TTTATTGAAACATCAGCAAAGACAAGACAGAGAGTGGAGGATGCTTTTTATACATTGGTGAGAGAGATCC
GACAATACAGATTGAAAAAAATCAGCAAAGAAGAAAAGACTCCTGGCTGTGTGAAAATTAAAAAATGCAT
TATAATGTAA

b)

Nucleotide Sequence (567 nt):

ATGACTGAATATAAACTTGTGGTAGTTGGAGCTGGTGGCGTAGGCAAGAGTGCCTTGACGATACAGCTAA
TTCAGAATCATTTTGTGGACGAATATGATCCAACAATAGAGGATTCCTACAGGAAGCAAGTAGTAATTGA
TGGAGAAACCTGTCTCTTGGATATTCTCGACACAGCAGGTCAAGAGGAGTACAGTGCAATGAGGGACCAG
TACATGAGGACTGGGGAGGGCTTTCTTTGTGTATTTGCCATAAATAATACTAAATCATTTGAAGATATTC
ACCATTATAGAGAACAAATTAAAAGAGTTAAGGACTCTGAAGATGTACCTATGGTCCTAGTAGGAAATAA
ATGTGATTTGCCTTCTAGAACAGTAGACACAAAACAGGCTCAGGACTTAGCAAGAAGTTATGGAATTCCT
TTTATTGAAACATCAGCAAAGACAAGACAGGGTGTTGATGATGCCTTCTATACATTAGTTCGAGAAATTC
GAAAACATAAAGAAAAGATGAGCAAAGATGGTAAAAAGAAGAAAAAGAAGTCAAAGACAAAGTGTGTAAT
TATGTAA

c)

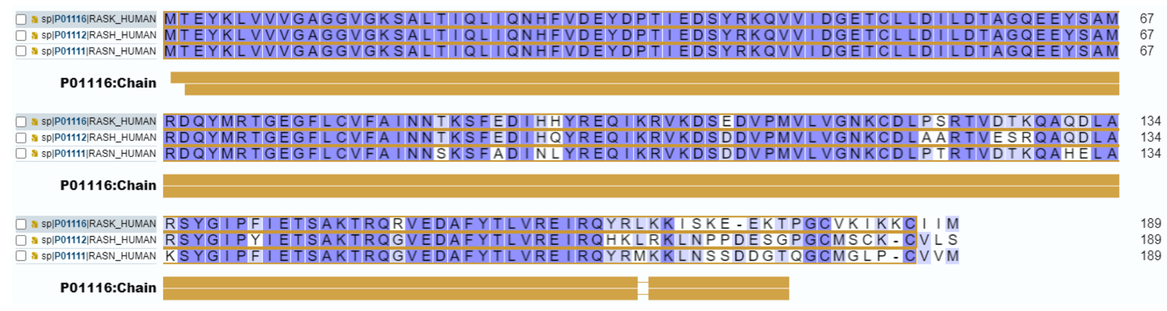

Figure 1: a) coding sequence of KRAS Isoform A. b) coding sequence of KRAS Isoform B. c) Alignment of KRAS, NRAS and HRAS

a)

b)

Percent Identity Matrix

sp|P01116|RASK_HUMAN	100.00%	87.23%	87.23%
sp|P01112|RASH_HUMAN	87.23%	100.00%	85.19%
sp|P01111|RASN_HUMAN	87.23%	85.19%	100.00%

c)

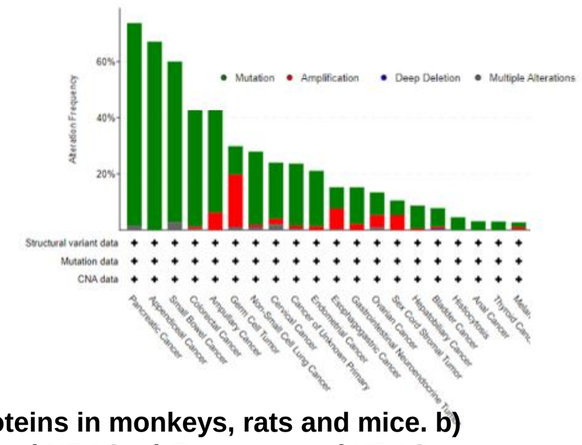

Figure 2: a) Identify and similarity of KRAS proteins in monkeys, rats and mice. b) Protein identify matrix between KRAS, HRAS and NRAS. c) Frequency of KRAS mutations in different type of cancer.

Discussion

- RAS genes have 3 isoforms – KRAS, HRAS and NRAS and show > 85% identify.
- KRAS contains 2 isoforms – Isoform KRAS 4A and KRAS 4B and they differ by one AA. KRAS 4B is the most common type of isoform.
- KRAS is highly homologous (> 98.9 %) across different species. (Monkey, Rat and Mice).
- Most common KRAS mutation is G12C and found in many cancers including NSCLC (non-small cell lung cancer) and CRC (colorectal cancer).
- G12C mutation activates the KRAS genes and sets to “on” position.
- Other rare KRAS mutations include G12S, G12V, G12D and G12A.
- Targeted therapies such as Adagrasib inhibits the KRAS G12C mutation.

References

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